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TRADITIONAL & KAMPO MEDICINE

Current situation and future issues with Kampo medicine: A survey of Japanese physicians

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Abstract

Aim: We aimed to elucidate the present clinical situation, future challenges, and potential need for diagnostic support software in Kampo medicine.

Methods: We conducted an online survey (comprising 28 questions) of Japanese physicians from the 23 to 24 January 2024 via MedPeer. The participants' responses were stratified according to their age distribution.

Results: We evaluated data for 652 respondents. The most common respondent age groups were those in their 30s and 40s. The most common specialty of respondents was internal medicine (43.7%). Only 33.7% of respondents had received education on Kampo medicine at university. Most respondents (90.0%) were not licensed as certified physicians or members of the Japan Society for Oriental Medicine. Most respondents (86.7%) currently prescribed Kampo formulas (current users); only 3.8% had no experience in prescribing Kampo formulas. The conditions for which Kampo formulas were frequently prescribed were muscle cramps, constipation, medically unexplained symptoms/menopausal syndrome, appetite loss/malnutrition, and fatigue. However, 48.3% of current users did not use traditional Kampo patterns (sho). The most frequent issues reported by respondents were the lack of scientific evidence and standardization of diagnostic methods in Kampo medicine. Respondents indicated the need for postgraduate education and implementation of diagnostic support software for using Kampo patterns by non-specialists in Kampo medicine.

Conclusion: There was a discrepancy between the high frequency of Kampo formula prescriptions and the low use of traditional Kampo patterns in Japan. Diagnostic support software is likely to become a useful method of improving appropriate Kampo formula prescriptions by non-specialists.

KEYWORDS
Kampo medicine, Kampo pattern (sho), medical education, online survey, software as a medical device (SaMD)

INTRODUCTION

Kampo medicine is a traditional Japanese medicine that involves the use of herbal formulas. The Japanese health insurance system covers 148 ethical Kampo formulas and 187 crude drugs. Therefore, many Japanese physicians prescribe Kampo formulas to patients [1]. Moreover, Japanese patients are interested in Kampo medicine as a therapeutic option [2].

Kampo formula prescriptions are based on the traditional theory and diagnostic methods of Kampo medicine. In 2001, the Japanese Model Core Curriculum for

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Medical University Education (first edition) started to include Kampo medicine [3]. However, many physicians in Japan still have no educational experience with or training in Kampo medicine. Therefore, it is unclear how Japanese physicians choose Kampo formulas to prescribe in daily clinical practice.

In recent years, there has been a digital transformation in medical care. Some studies have reported that diagnostic support software increases the diagnostic accuracy of physicians and residents [4–6]. Therefore, implementing diagnostic support systems may improve the accuracy of Kampo formula prescriptions and enhance the implementation of appropriate Kampo therapy by Japanese physicians. However, there are no studies on Japanese physicians' attitudes toward diagnostic support systems in Kampo medicine.

We conducted an online survey of Japanese physicians to elucidate the present clinical situation, future challenges, and potential need for tools that support the use of Kampo medicine.

METHODS

Study design

We conducted an online survey of Japanese physicians from the 23 to 24 January 2024 using MedPeer (https://medpeer.co.jp/). We used a questionnaire to obtain information on the current clinical use and diagnostic approach of Kampo medicine in Japan. Furthermore, we evaluated the key issues, future expectations, and value of the diagnostic support system.

The need for ethical review was waived by the ethical committee at Fukushima Medical University (REC2023-149) because the Ethical Guidelines for Medical and Health Research Involving Human Subjects were not applicable to the present study [7]. The study protocol was registered at the UMIN Clinical Trials Registry (UMIN000053359).

Study population

We recruited licensed Japanese physicians registered at MedPeer who agreed with the study purpose. Physicians in junior residency programs were excluded from the study. A sample size calculation was performed using the following formula: $n = Z^2P(1 - P)/d^2$, with $Z = 1.96$ for a confidence level of 95%, $P = 0.5$ for percentage of population probability, and $d = 0.05$ for the margin of error [8]. According to the formula, the required sample size was 384. The sample size was multiplied by a predicted design effect of 2 to account for convenience sampling and the use of an online survey. Hence, the optimal sample size for each region was set as 600 (384×1.5). The participants' responses were almost stratified according to the age distribution of Japanese physicians [9].

Questionnaire

The questionnaire contained 28 questions presented in single- and multiple-choice, Likert scale, and fill-in formats. The original questionnaire was written in Japanese (Table S1). The English version of the questionnaire is presented in Table S2. The participants self-completed the questionnaires. After all the participants had responded, MedPeer sent us the data.

Data collection

Respondents' clinical fields were grouped into broad categories as follows: (1) general internal medicine, geriatrics and gerontology, cardiology, gastroenterological medicine, respiratory medicine, infectious diseases, neurology, metabolism and endocrinology, nephrology and dialysis, collagen diseases, rheumatology, allergology, hematology, and medical oncology were categorized as internal medicine; (2) general surgery, cardiovascular surgery, vascular surgery, gastroenterological surgery, respiratory surgery, neurosurgery, breast and endocrine surgery, urology, dermatology, ophthalmology, otorhinolaryngology, and plastic surgery were categorized as surgery; (3) pediatrics; (4) obstetrics and gynecology; (5) orthopedics, sports medicine, and rehabilitation were categorized as orthopedics; (6) psychiatry and psychosomatic medicine were categorized as psychiatry; (7) general practice, home medical care, family medicine, radiology, anesthesiology, palliative care, emergency medicine, intensive care, industrial medicine, medical checkup/preventive medicine, Kampo medicine, integrative and alternative medicine, esthetic and antiaging medicine, and others were categorized as others.

We defined respondents who currently prescribed Kampo medicine (current users) as those who answered 1 to question 1. We defined respondents with undergraduate university education in Kampo medicine as those who answered 1 to question 27. We defined specialists in Kampo medicine as respondents who answered 1 (licensed as a certified physician of the Japan Society for Oriental Medicine (JSOM)) or 2 (licensed as a member of the JSOM) to question 26.

Data representation

All values are expressed as means.

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TABLE 1 Respondents' characteristics.

	All respondents		Current users		Past users		Non-users	
	n	%	*n*	%	*n*	%	*n*	%
Total								
	652	100.0	565	100.0	62	100.0	25	100.0
Age								
20s	30	4.6	23	4.1	3	4.8	4	16.0
30s	140	21.5	125	22.1	10	16.1	5	20.0
40s	140	21.5	129	22.8	8	12.9	3	12.0
50s	138	21.2	123	21.8	10	16.1	5	20.0
60s	115	17.6	95	16.8	16	25.8	4	16.0
70s	82	12.6	66	11.7	12	19.4	4	16.0
80s	7	1.1	4	0.7	3	4.8	0	0.0
90s	0	0.0	0	0.0	0	0.0	0	0.0
Clinical department								
Internal medicine	285	43.7	258	45.7	21	33.9	6	24.0
Surgery	129	19.8	105	18.6	16	25.8	8	32.0
Pediatrics	45	6.9	38	6.7	5	8.1	2	8.0
Obstetrics and gynecology	16	2.5	16	2.8	0	0.0	0	0.0
Orthopedics	34	5.2	33	5.8	1	1.6	0	0.0
Psychology	58	8.9	52	9.2	4	6.5	2	8.0
Others	85	13.0	63	11.2	15	24.2	7	28.0
Undergraduate education in Kampo medicine								
Yes	220	33.7	204	36.1	11	17.7	5	20.0
No	311	47.7	258	45.7	41	66.1	12	48.0
Do not remember	121	18.6	103	18.2	10	16.1	8	32.0
Board-certified physician or member of the JSOM								
Physician	45	6.9	41	7.3	2	3.2	2	8.0
Member	20	3.1	19	3.4	1	1.6	0	0.0
None	587	90.0	505	89.4	59	95.2	23	92.0
Type of facility								
Doctor's office, clinic	167	25.6	140	24.8	20	32.3	7	28.0
University hospital	114	17.5	104	18.4	5	8.1	5	20.0
National and public hospital	122	18.7	106	18.8	11	17.7	5	20.0
Hospital other than university, national, or public hospital	240	36.8	212	37.5	20	32.3	8	32.0
Others	9	1.4	3	0.5	6	9.7	0	0.0

Note: Current users, respondents who currently prescribe Kampo formulas; past users, respondents who have prescribed Kampo formulas in the past but do not currently prescribe Kampo formulas; non-users, respondents who have never prescribed Kampo formulas.
Abbreviation: JSOM, Japan Society for Oriental Medicine.

RESULTS

Respondents' characteristics

Figure S1 shows the study flowchart. There was a total of 685 respondents. Excluding junior residents, we finally assessed 652 respondents (Table 1). All participant responses are presented in Tables S3 and S4. The most common respondent age groups were the 30s (21.5%) and 40s (21.5%). The most common specialty was internal medicine (43.7% of respondents) and 73.0% of respondents worked at a hospital. Only 33.7% of respondents had received undergraduate education in Kampo medicine. In addition, 90.0% of respondents were not certified physicians or members of the JSOM.

Frequency of Kampo formula prescription by Japanese physicians

There were 565 respondents (86.7%) who currently prescribed Kampo formulas (current users), while 62 respondents (9.5%) had previously prescribed

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Kampo formulas (past users). In contrast, only 25 respondents (3.8%) had never prescribed Kampo formulas (non-users). The distributions of age, undergraduate education, Kampo medicine specialist status, and type of health facility for current users were similar to the distributions for all respondents. All respondents who specialized in obstetrics and gynecology were current users. The proportion of respondents with undergraduate education or specialist licenses in Kampo medicine was higher among current users than among non-users.

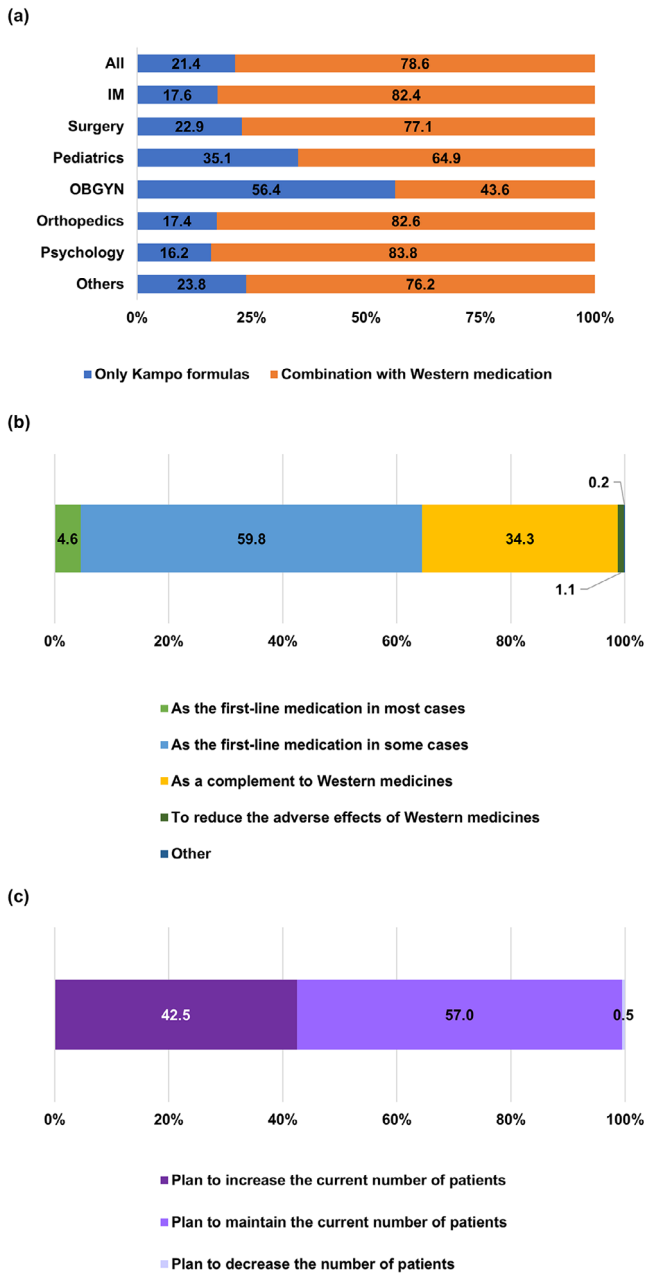

FIGURE 1 Current clinical situation and future intentions regarding the use of Kampo medicine among current users. The total number of respondents was 565 (current users). (a) Distribution of single or combination prescriptions of Kampo and Western medicines to patients. IM, internal medicine; OBGYN, obstetrics and gynecology. (b) Kampo formulas as the first-line therapy. (c) Future intentions regarding Kampo formula prescriptions.

Combination therapy of Kampo formulas and Western medicines

We asked current users about the clinical situation of prescribing Kampo formulas. Kampo formulas were prescribed to 14.0% of patients (Table S3, question 5). In 78.6% of cases, current users combined Kampo formulas and Western medication (Figure 1a). Higher proportions of respondents who specialized in obstetrics–gynecology and pediatrics administered only Kampo formula prescriptions (56.4% and 35.1%, respectively). The average number of Kampo formula types prescribed by current users was 8.6, and this number increased with respondent age (Figure S2).

We asked current users about the role of Kampo formulas in clinical practice. Kampo formulas were considered the first-line therapy for patients with specific conditions by 59.8% of respondents (Figure 1b). In contrast, only 4.6% of respondents considered Kampo formulas as their first choice for most patients. The most common reasons for Kampo formula prescriptions were 'insufficient response to Western medicine' and 'patient's request' (277 and 218 respondents, respectively) (Figure S3).

Evaluation and future prospects of Kampo medicine by Japanese physicians

The Likert scale response evaluations of Kampo therapy are shown in Figure S4. Most current users evaluated Kampo formulas favorably on therapeutic efficacy, patient satisfaction, and cost-effectiveness. Figure 1c shows current users' intentions regarding the future prescription of Kampo formulas. Most current users intended to maintain the present number of patients (57.0%) or increase the number of patients treated with Kampo formulas (42.5%). In contrast, few current users wanted to reduce the number of patients prescribed Kampo formulas (Figure 1c).

Diseases, disorders, and symptoms targeted by Kampo formulas

Table 2 lists the diseases, disorders, and symptoms that current users treated using Kampo formulas. The conditions most frequently treated with Kampo formulas were: muscle cramps, constipation, medically unexplained symptoms (MUS)/menopausal syndrome, appetite loss/malnutrition, and fatigue (260, 214, 169, 163, and 159 respondents, respectively). Biliary and pancreatic diseases, chronic hepatitis, and

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TABLE 2 List of diseases, disorders, and symptoms treated with Kampo formulas in Japan.

Ranking	Diseases, disorders, and symptoms	Total respondents *n*	Number of corresponding patients selected by respondents (multiple answers allowed)						
			Children (< 15 years old)	Male (16–39 years old)	Female (16–39 years old)	Male (40–59 years old)	Female (40–59 years old)	Male (≥ 60 years old)	Female (≥ 60 years old)
1	Muscle cramps	260	5	11	19	75	92	184	193
2	Constipation	214	12	6	21	32	56	137	156
3	Medically unexplained symptoms, menopausal syndrome	169	4	3	13	22	115	31	88
4	Appetite loss, malnutrition	163	8	10	13	30	42	113	130
5	Fatigue, malaise	159	9	17	23	51	66	91	108
6	Dementia and its behavioral and psychological symptoms	120	1	0	1	3	4	99	100
7	Acute upper respiratory inflammation	108	19	42	55	60	67	60	62
8	Edema	101	2	3	18	14	41	42	71
9	Cough and phlegm	94	10	19	22	44	47	57	54
10	Dysautonomia	73	6	10	21	18	46	13	39
11	Vertigo and dizziness	63	2	4	10	18	32	25	35
12	Ileus	61	5	4	5	14	21	45	40
13	Irregular menstruation/ dysmenorrhea	52	1	0	27	3	33	2	4
14	Allergic rhinitis	51	6	21	26	29	28	22	23
15	Gastritis	47	2	4	9	18	18	28	19
16	Headache	43	7	5	16	13	24	12	10
17	Urinary disorder	41	1	2	1	4	3	31	21
18	Postoperative complaints	40	2	3	4	13	13	27	28
19	Diarrhea	35	9	5	9	11	9	17	13
20	Irritable bowel syndrome	34	13	8	11	9	11	6	7
21	Frequent urination	31	1	1	2	1	4	23	18
22	Back pain	29	0	1	1	3	8	19	19
23	Joint pain/knee osteoarthritis	28	0	0	1	5	7	16	23
24	Other	27	7	4	7	3	7	12	12
25	Gastroesophageal reflux disease	25	5	2	2	8	7	9	9
26	Eczema, skin pruritus	22	1	3	7	6	9	10	9
27	Acute and chronic bronchitis	20	0	4	4	9	9	13	10
27	Cystitis	20	0	1	3	4	8	5	17
29	Asthma	10	3	0	1	2	5	2	2
29	Atopic dermatitis	10	1	3	3	4	1	1	2

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TABLE 2 (Continued)

Ranking	Diseases, disorders, and symptoms	Total respondents	Number of corresponding patients selected by respondents (multiple answers allowed)						
		n	Children (< 15 years old)	Male (16–39 years old)	Female (16–39 years old)	Male (40–59 years old)	Female (40–59 years old)	Male (≥ 60 years old)	Female (≥ 60 years old)
31	Sequelae of cerebrovascular disease	9	0	0	0	2	1	7	6
32	Gastrointestinal ulcer	8	1	1	1	0	2	3	2
33	Chronic obstructive pulmonary disease	7	0	1	1	1	0	6	1
33	Rheumatoid arthritis	7	0	0	0	2	2	3	3
35	Nephritis, nephrotic syndrome	6	0	0	2	2	2	3	1
35	Diabetes and its complications	6	0	0	1	3	3	3	4
37	Cerebrovascular disease	5	1	0	0	0	0	3	4
38	Hypertension	4	0	0	1	0	1	1	2
38	Chronic hepatitis	4	0	1	1	0	2	1	0
40	Biliary and pancreatic diseases	2	0	0	0	0	0	2	1

Note: The total number of respondents was 565 (current users). Respondent were allowed to choose any number of patient age and sex groups for up to five diseases, disorders, or symptoms. Others included the following conditions: common cold, shoulder stiffness, crying at night, stomatitis, pharyngolaryngeal paresthesia, peripheral neuropathy, neurological symptoms, chronic subdural hematoma, delirium associated with dementia, orthostatic dysregulation, insomnia, cramps, vascular insufficiency of the lower limbs, perianal abscess, hemorrhoids, male infertility, post-surgical ileus prevention, rosacea, and acne.

hypertension were less commonly treated with Kampo medicine by current users.

Table 2 also presents the age and sex of patients prescribed Kampo medicine. Overall, Kampo formulas were frequently prescribed to older adults, but were frequently prescribed to younger patients for upper respiratory tract infections, allergic diseases, headaches, and functional gastrointestinal diseases.

Use of traditional Kampo medicine patterns

We evaluated the number of current users using traditional patterns in Kampo medicine ('sho' in Japanese). Approximately half of current users (48.3%) did not consider Kampo patterns when prescribing Kampo formulas (Figure 2, Table S5). The proportion of current users using only Western medicine diagnoses was lower for obstetrician–gynecologists and Kampo specialists (25.0% and 16.7%, respectively). Few physicians referred only to traditional patterns when prescribing Kampo formulas, regardless of age group, clinical department, educational experience, or specialization in Kampo medicine.

Our results also showed that Japanese physicians thought that being unaware of traditional Kampo patterns could lead to disadvantages, such as reduced therapeutic effects, patient dissatisfaction, and adverse effects (Figure 3, Table S6). The proportion of respondents who mentioned patient dissatisfaction as a disadvantage of lack of awareness of traditional Kampo patterns was higher among pediatricians and obstetrician–gynecologists than among physicians from other specialties.

Undergraduate education in Kampo medicine

Undergraduate education in Kampo medicine was considered favorably ('very meaningful' or 'meaningful') by 44.2% of current users (Figure S5a). Aspects of education considered important by current users were 'typical indications for Kampo formulas', 'basic theories of Kampo medicine (traditional components of sho)', 'physical examination methods (four examinations in Kampo medicine)', and 'adverse effects of Kampo formulas' (Figure S5b).

Reasons for avoiding Kampo formula prescriptions and future intentions of past users and non-users

Figure 4 summarizes the reasons that past users and non-users gave for not prescribing Kampo formulas. The most important reason was 'difficulty using Kampo

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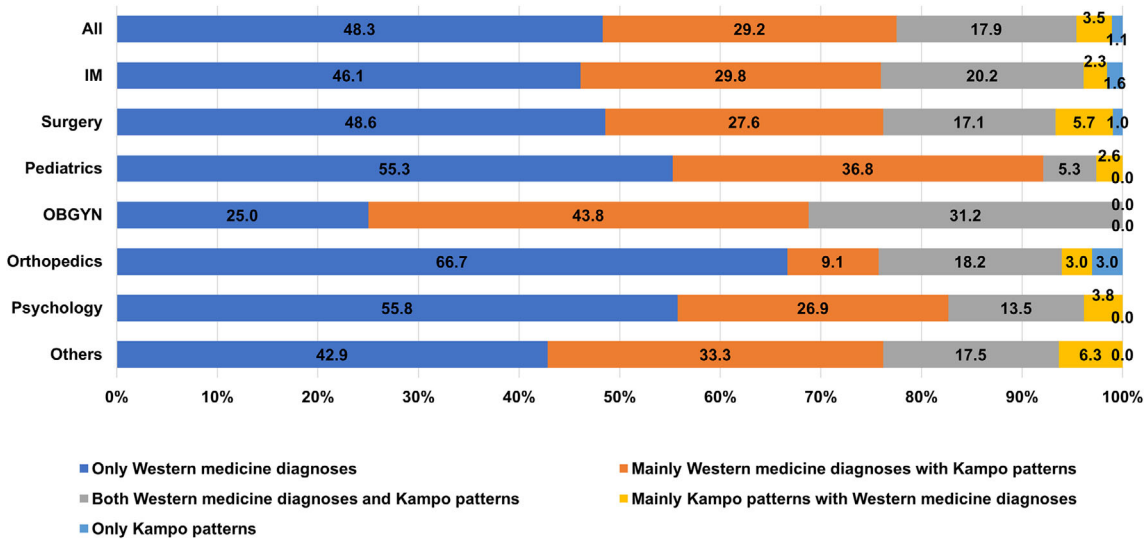

FIGURE 2 Use of traditional patterns (sho) to select Kampo formulas by current users. The total number of respondents was 565 (current users). IM, internal medicine; OBGYN, obstetrics and gynecology.

formulas' (37 respondents); other common reasons were 'Western medicines are sufficient for treatment', 'patients did not favor Kampo formulas', and 'insufficient evidence' (Figure 4b). However, 80.5% of past users and non-users expressed a desire to prescribe Kampo formulas in the future (Figure 4c).

Future expectations and key issues in Kampo medicine

Figure 5a shows responses from all participants about the future roles of Kampo medicine. The most common reasons for using Kampo medicine were for MUS and for 'curing diseases that were difficult to treat using Western medicine'; other reasons were to 'support and complement Western medicine', 'improvement of systemic symptoms and constitution', 'medical care for older adults', and 'improvement of quality of life'.

The most common responses from all participants regarding issues in Kampo medicine that needed addressing in the future were 'gathering scientific evidence for the efficacy of Kampo medicine' and 'standardizing the diagnostic method of Kampo medicine' (170 and 94 respondents, respectively) (Figure 5b). The next most identified issues were the interpretation and objectification of traditional theory and diagnostic methods.

Anticipated solutions for promoting appropriate use of Kampo medicine by non-specialists

Figure 6 and Table S7 show all respondents' answers regarding solutions for physicians who do not specialize in Kampo medicine to identify traditional Kampo patterns. More respondents mentioned postgraduate education (55.2%) and diagnostic support software in Kampo medicine (42.6%) than undergraduate education (360, 278, and 214 respondents, respectively). Approximately 40% of respondents mentioned diagnostic support software, regardless of age group, clinical department, educational experience, or specialization in Kampo medicine.

DISCUSSION

This is the first online survey of Japanese physicians' attitudes toward the current clinical situation, future issues, and solutions for promoting the use of appropriate Kampo formula prescriptions by nonspecialists. The main findings were as follows. First, most Japanese physicians frequently prescribe Kampo formulas for patients with various conditions, regardless of age group, clinical department, educational experience, and specialization in Kampo medicine. Second, approximately half of the Japanese physicians surveyed had difficulty using traditional Kampo patterns (sho). Each physician treated patients with a limited number of conditions using fewer than 10 types of Kampo formulas. However, the physicians realized that being unaware of traditional Kampo patterns could lead to disadvantages, such as insufficient therapeutic effects of Kampo formulas. Third, regardless of the current use of Kampo formulas, most respondents intended to use Kampo formulas in the future. Fourth, Japanese physicians believed that accumulating evidence and standardizing diagnostic methods in Kampo medicine were the main issues that needed to be addressed. Fifth, enhancing postgraduate education and developing

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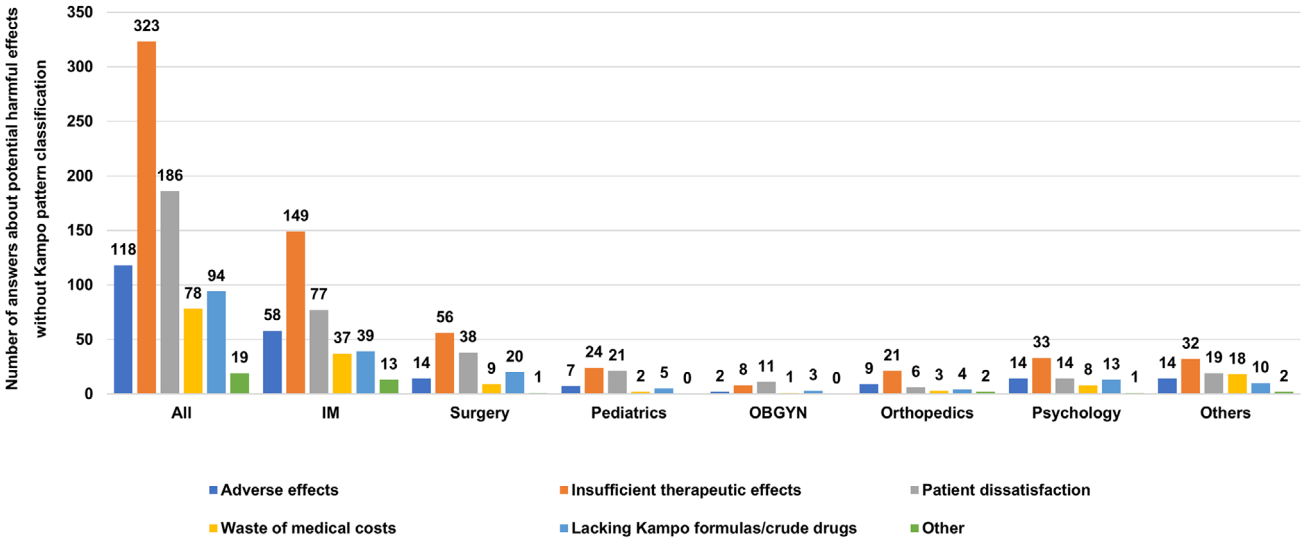

FIGURE 3 Potential harmful effects of Kampo medical treatment without pattern classification. The total number of respondents was 565 (current users). Respondents were allowed to choose multiple answers. IM, internal medicine; OBGYN, obstetrics and gynecology.

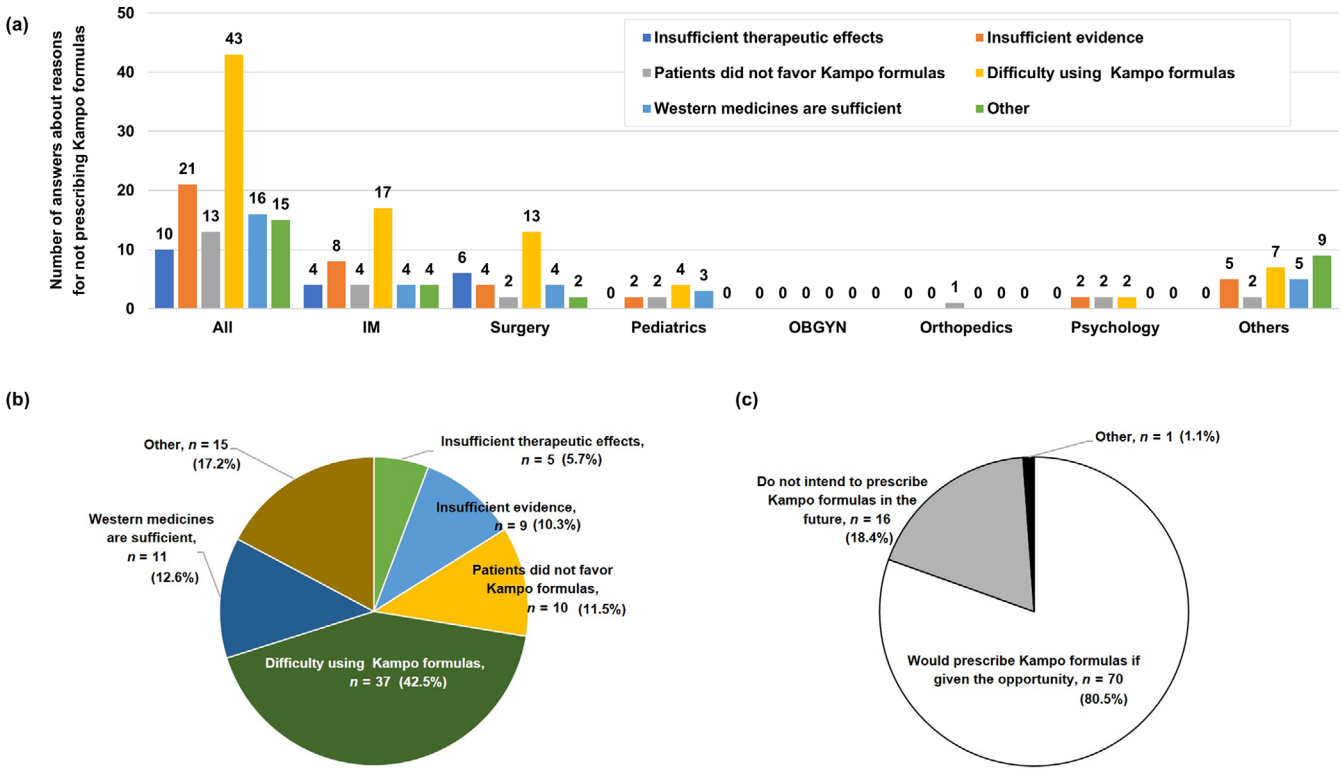

FIGURE 4 Reported reasons for avoiding the prescription of Kampo formulas and future intentions of past users and non-users. The total number of respondents was 87 (past users and non-users). (a) Reasons for avoiding prescribing Kampo formulas. Respondents were allowed to choose multiple answers. IM, internal medicine; OBGYN, obstetrics and gynecology. (b) Top reasons for avoiding prescribing Kampo formulas. (c) Position on the future prescription of Kampo formulas.

diagnostic support software were identified as potential ways of promoting appropriate Kampo treatment using pattern identification by non-specialists.

Our study showed that most Japanese physicians had experience in prescribing Kampo formulas, as shown in previous reports [10–12]. Moreover, Kampo formulas were prescribed for diverse conditions. These results indicated that Japanese physicians and patients have been familiar with Kampo medicine for many years. However, our results also clarified that Japanese

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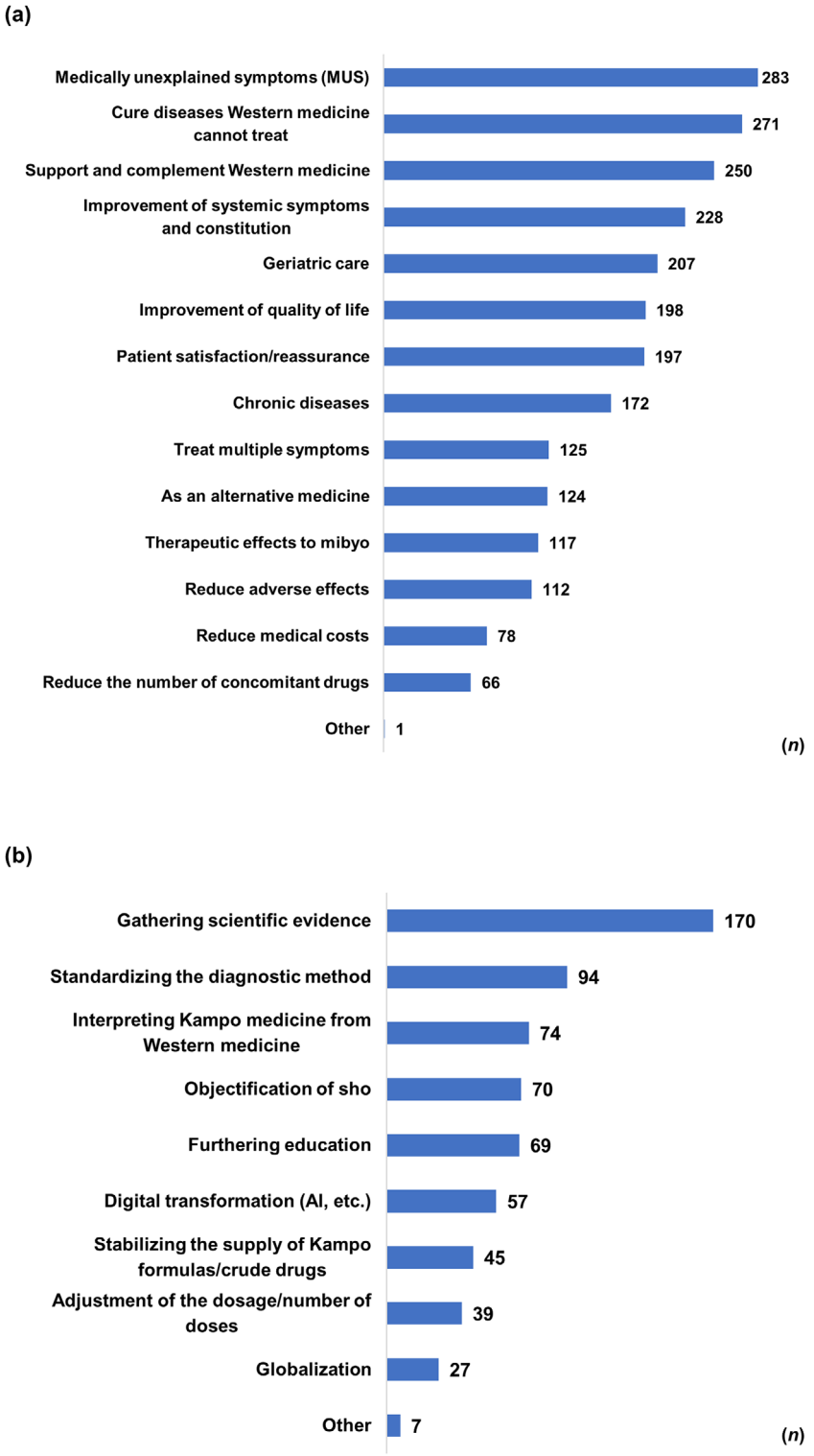

FIGURE 5 Future roles and key issues regarding Kampo medicine. The total number of respondents was 652 (current users, past users, and non-users). (a) Future roles of Kampo medicine. Respondents were allowed to choose multiple answers. (b) Top issues to be addressed in Kampo medicine.

physicians remain unable to perform Kampo treatment by pattern identification. The proportion of physicians practicing Kampo treatment by pattern identification has not increased since 2011 [10]. However, current Japanese physicians have realized the potential problems caused by ignorance of traditional Kampo patterns. These results highlight the need to promote post-graduate education on traditional Kampo pattern identification in clinical settings for Japanese physicians.

Surprisingly, we observed favorable responses regarding the present use of Kampo formulas. Past users and non-users reported that difficulty using Kampo formulas was the most common reason that

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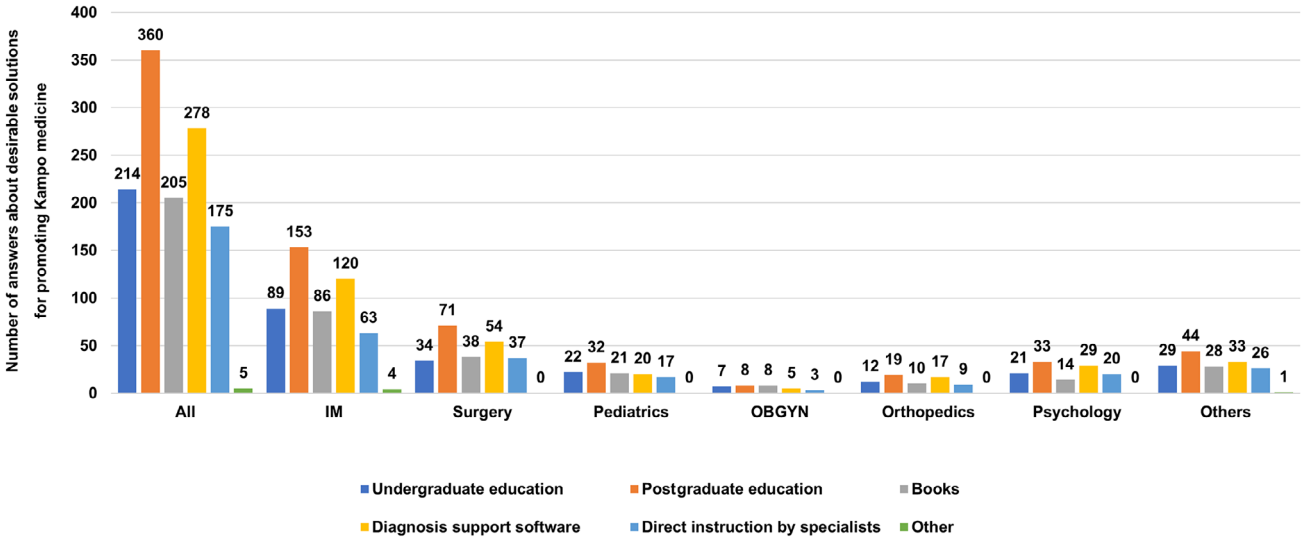

FIGURE 6 Anticipated solutions for promoting Kampo medicine. The total number of respondents was 652 (current users, past users, and non-users). Respondents were allowed to choose multiple answers. IM, internal medicine; OBGYN, obstetrics and gynecology.

they avoided prescribing Kampo medicine. Furthermore, the treatment targets of Kampo medicine included pathological conditions that are difficult to treat with Western medicine, such as MUS and fatigue [13–15]. Therefore, these results suggest the need for diagnostic support tools in Kampo medicine to improve treatment using pattern identification by non-specialists in Kampo medicine.

We showed that postgraduate education and diagnostic support software are candidate solutions to the abovementioned problems. Recent studies have reported that clinical diagnostic support systems have educational effects on medical students, residents, and physicians [5, 16, 17]. Therefore, diagnostic support software in Kampo medicine that can provide educational feedback to users is likely to be useful for postgraduate education.

Our study had several limitations. First, we only evaluated Japanese physicians registered in the MedPeer database. Second, we could not demonstrate causality because of the cross-sectional study design.

In conclusion, we found that Japanese physicians frequently use Kampo formulas but do not often use traditional Kampo patterns (sho). The implementation of diagnostic software is a potential approach for resolving this inconsistency and promoting appropriate Kampo formula prescription by non-specialists. Further exploration of the use of Japanese Kampo medicine is warranted.

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CONFLICT OF INTEREST STATEMENT

Kazushi Uneda received research grant support from TSUMURA & CO. Tetsuhiro Yoshino was employed at Keio University for collaborative research with TSUMURA & CO. Tatsuya Nogami received lecture fees from TSUMURA & CO. and Kracie, Ltd. The authors declare that the conduct of this research was unaffected by any commercial or financial relationships that could be construed as potential conflicts of interest.

APPROVAL OF THE RESEARCH PROTOCOL BY AN INSTITUTIONAL REVIEWER BOARD AND THE APPROVAL NUMBER

The need for ethical review was waived by the ethical committee at Fukushima Medical University (REC2023-149) because the Ethical Guidelines for Medical and Health Research Involving Human Subjects were deemed not applicable to the present study.

INFORMED CONSENT

N/A.

REGISTRY AND THE REGISTRATION NO. OF THE STUDY/TRIAL

N/A.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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